CONTROLLING INFORMATION RELIABILITY AT BORDER POINTS

Babaqulov Ilxomjon Xaliqulovich
Chief Specialist at the "Digital Government Project Management Center"
ilxomb@gmail.com
Rustamova Moxichexra Yaxshiboyevna
Senior Lecturer at Tashkent University of Information Technologies named after Muhammad al-Khwarizmi
mohimrustamova83@gmail.com
Abilov Dilshod Bekzod o‘g‘li
Student at Tashkent University of Information Technologies named after Muhammad al-Khwarizmi
abilovdilshod55@gmail.com
Abdusattorov Samandar O‘tkirjon o‘g‘li
Student at Tashkent University of Information Technologies named after Muhammad al-Khwarizmi
samandar180807@gmail.com
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Annotation

This article examines modern methods of controlling the reliability of information at border checkpoints. A multi-stage information reliability control model has been developed to address problems such as data errors, false documents, and interdepartmental information conflicts in existing systems. The model is based on cross-validation algorithms, a reliability index I(x), and an automatic classification mechanism, through which incoming data is automatically evaluated based on the level of reliability. The results of the study allow us to reduce data errors by 78 percent and move the control process from a reactive approach to proactive management.

Keywords: information reliability, border control, cross-validation, Reliability Index, VIS-limit, data quality, automatic control, digital integration.

Introduction

Modern border checkpoints process hundreds of cargo units at a time, generating thousands of data records in the process. In this context, ensuring the reliability of information is becoming not only a technical but also a strategic management issue. In border control, information reliability refers to the completeness, accuracy, consistency, and compliance with regulatory requirements of incoming data.

Practice shows that 15–25 percent of the information processed at border crossings contains various errors, inconsistencies, or omissions. This leads to the adoption of incorrect control decisions, the release of dangerous goods, or, conversely, the unjustified detention of safe goods. In addition, the problem of falsified documents remains one of the most pressing issues of border control: according to the World Customs Organization (WCO), 3–7% of goods crossing borders are found to contain falsified documents.

The introduction of the VIS-Border Information System in the Republic of Uzbekistan was an important step towards digitization of information exchange processes [1, 2]. However, in the existing system, mechanisms for automatic control of information reliability are not sufficiently developed. Verifying the accuracy of incoming data often relies on manual operations by inspectors, which increases the impact of the human factor and reduces control effectiveness [3].

The purpose of this study is to develop a scientifically based model of automatic control of information reliability at border points and evaluate its effectiveness on the example of the Vis-border system.

Research tasks:

identify the main factors affecting the reliability of information at border crossings;

develop a mathematical model for calculating the reliability index;

design an automatic control mechanism based on cross-validation;

evaluate the practical effectiveness of the proposed model.

Methodology

The study used a two-stage mixed methodology. The theoretical stage analyzed more than 45 scientific sources published between 2015 and 2025 on information reliability, data quality, and cross-validation algorithms. In the practical stage, the VIS-Border system database for the period 2023–2025 was used as an empirical source.

A dataset of 11,400 cargo units was compiled for analysis from 4 main border checkpoints in Uzbekistan - Uzbekistan-Kazakhstan, Uzbekistan-Tajikistan. The composition of the collection includes: cargo documents and certificates, country of origin and route data, results of a previous examination, laboratory conclusions and historical risk indicators.

Mathematical model of information reliability

To assess the reliability of information, the multi-criteria reliability index I(x) is calculated using the following formula:

$$I(x)=\sum_{i=1}^{n} w_i \cdot K_i(x)$$

Here I(x) is the overall reliability index (from 0 to 1), w_i is the weighting coefficient of the i-th criterion ($\sum w_i = 1$), and $K_i(x)$ is the reliability value of the i-th criterion.

The weight coefficients were determined by industry experts based on the AHP (Analytic Hierarchy Process) method and satisfied the condition of consistency ratio CR < 0.10. The main criteria and their weights are presented in Table 1.1-jadval. Ishonchlilik mezonlari va og‘irlik koeffitsientlari

Criterion	**Mark**	**Weight (w_i)**
Document completeness	K_1	0.25
Data consistency	K_2	0.20
Historical data compatibility	K_3	0.20
Compliance with regulatory requirements	K_4	0.20
Source reliability	K_5	0.15

Based on reliability index values, loads are classified into three categories: I(x) ≥ 0.80 — high reliability (green corridor), 0.50 ≤ I(x) < 0.80 — medium reliability (yellow corridor), i(x) < 0.50 — low reliability (Red Corridor).

Cross-validation mechanism

Stratified K-Fold cross-validation was used to control the reliability of the information. K was set to 10, meaning the dataset was divided into 10 equal parts, and 9 parts were used for training and 1 part for validation in each iteration. The stratified approach allowed us to maintain the category ratio in each fold, ensuring adequate representation of rare but high-risk categories.

In model training, methods for classification based on the CART decision tree algorithm, probability calculation by logistic regression, and determination of Criterion weights using the

Random Forest algorithm were used, and all calculations were carried out in the Python 3.11 environment through the scikit-learn library. The results were verified by two independent researchers.

Comparison methodology

The study included a control group that used the traditional manual inspection method and an experimental group that used the proposed model. The two groups were given a comparative assessment of four criteria — accuracy, time consumption, proportion of false positives, and degree of anomaly detection.

Results

State of information reliability in an existing system

Analysis of 11,400 cargo units using the traditional inspection method in the control group showed the following results. The data error rate was 17.3 percent, meaning that one in every 6 cargo units had a data error. The composition of the identified errors was distributed as follows: document incompleteness — 38 percent, data inconsistency — 29 percent, compliance with regulatory requirements — 21 percent, source reliability — 12 percent.

The average inspection time was 43 minutes. False document cases accounted for 4.2 percent of all inspection results. The anomaly detection rate was 61%, meaning that 39% of dangerous cases remained undetectable in the traditional way.

Results of the proposed model

When the cross-validation-based reliability control model was applied to the experimental group, the following results were obtained. The data set was distributed by class as follows: high reliability (green) — 8,208 units (72.0%), medium reliability (yellow) — 2,280 units (20.0%), low reliability (red) — 912 units (8.0%).

The Cross-validation model reduced data error from 17.3 percent to 3.8 percent, representing a 78 percent decrease. The inspection time was reduced from 43 minutes to 7 minutes, 84 percent. The rate of detection of forged documents increased from 61 percent to 94 percent. The model independently detected 23 cases of forged documents that were not detected at all by the previous system.

Further investigations of the 68 low-confidence cases identified: 54 (79.4%) were confirmed to be truly dangerous, 10 (14.7%) were placed under further observation, and 4 (5.9%) were deemed safe. This proves that the level of false positive results is very low.

Results of comparative analysis

A comparative analysis of the two systems is presented in Table 2.

Table 2. Traditional system and proposed model comparison results

Criterion	**Traditional system**	**Proposed model**	**Improvement**
Ma’lumotlar xatoligi	17.3%	3.8%	78.0% decreased
Tekshiruv vaqti	43 daqiqa	7 daqiqa	84.0% shrunk
Anomaliya aniqlash	61%	94%	33 points increased
O'tkazib yuborilgan xavf	Mavjud	Minimal	23 holat aniqlandi
Soxta musbat natija	8.1%	5.9%	2.2 points decreased

The system efficiency, calculated based on the integral efficiency model $E = \Sigma w_i \cdot M_i$, was $E = 0.81$, which represents a high level. In the classical risk model, however, $r = 1\text{-}E = 0.19$ showed lower risk levels.

Discussion

Scientific significance of the results

The results obtained showed that cross-validation algorithms can be highly effective in controlling information reliability in border control systems. A 78% reduction in data errors is one of the best results reported in the scientific literature in this field [4, 5]. A study by Wang et al. (2022) found that ML algorithms achieved an average error reduction of 65–72 percent in food control systems [5]. Our model performed 6–13 points higher than this figure, which is explained by the fact that the Stratified K-Fold method effectively solves the problem of data imbalance.

In a study by Gao, Levi, and Renegar (2022), an ensemble model was used to achieve 76 percent accuracy in risk assessment [4]. The achievement of a 94 percent anomaly detection rate in our model was ensured through the joint application of Random Forest and logistic regression.

Importance of Stratified K-Fold method

Stratified cross-validation gave particularly important results. In border control, low-reliability category loads (8.0%) are a rare but most critical category. In the simple K-Fold method, this category may not have been adequately represented. In the stratified approach, however, the ratio of categories was maintained in each fold, which significantly increased the model's ability to detect rare but dangerous cases.

Limitations and future research

There are also some limitations to the study. First, the model was tested based on data from four border checkpoints, and it is necessary to cover all checkpoints for a large-scale test. Second, cross-validation algorithms require additional computational resources to run in real time, which can be a problem at points with limited technical capabilities. Third, the model data is only from border crossings in Uzbekistan, and additional research is needed to test its international generalizability.

Future research aims to test the model on a large scale at all border checkpoints in Uzbekistan, further increase accuracy through integration with neural networks, and bring it up to international WCO as well as OIE standards.

Conclusion

This study developed a new model for controlling the reliability of information at border checkpoints based on cross-validation algorithms and tested it on the example of the VIS-Border system.

The results of the study showed that the proposed model reduced the data error rate from 17.3 percent to 3.8 percent, a reduction of 78 percent. The review time was reduced from 43 minutes to 7 minutes — an 84 percent reduction. The rate of anomaly detection increased from 61 percent to 94 percent, and 23 cases of fraudulent documents that were not detected at all by the traditional system were independently identified. Automatic classification of loads based on the reliability index I(x) made it possible to target the distribution of control resources.

These results form the technological basis for the transfer of the border control system from a traditional reactive approach to a modern proactive control system. The Model not only increases information reliability, but also plays an important role in ensuring biological and economic security.

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